(Material & Manufacutring Technology International Conferance & Exibition)
Poster Presentation Competiton
Predicitve Maintenance in furnances
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Poster Presentation Competiton
Predicitve Maintenance in furnances

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PREDICTIVE MAINTENANCE IN FURNACE

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Introduction

Maintenance is essential in the process industry to reduce equipment failures, downtime, and production losses. AI-based predictive maintenance enables continuous monitoring of furnace parameters such as three-phase current, power consumption, and fan operation. By detecting abnormal patterns and predicting possible faults at an early stage, the system helps technicians take timely action, improve furnace reliability, reduce maintenance costs, and ensure stable operation.



Introduction

BLAST FURNACE

↓

High Temperature + Continuous Operation

↓

Failure → Downtime + Energy Loss + Cost + Safety Risk

↓

Traditional Maintenance → Limited Early Detection

↓

Sensors + IoT + Data Analytics + AI/ML

↓

PREDICTIVE MAINTENANCE

↓

Early Fault Detection → Reduced Downtime → Improved Reliability

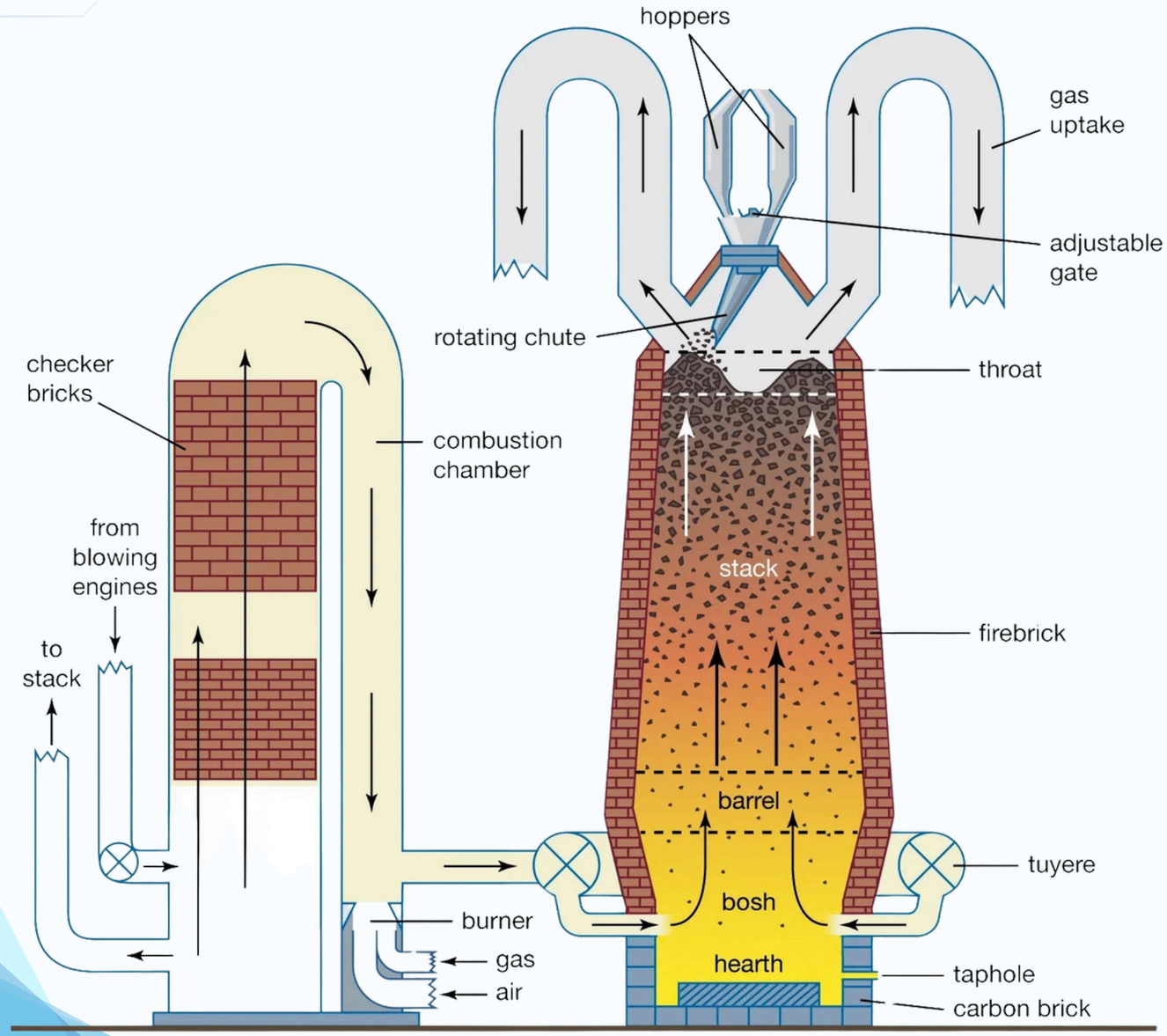
What Exactly Is a Blast Furnace?

A blast furnace is a vertical shaft furnace used for producing molten iron from iron-bearing materials.

Iron Ore / Sinter / Pellets
+
Coke
+
Fluxes
+
Hot Blast
↓
BLAST FURNACE
↓
Hot Metal + Slag + Blast Furnace Gas

Important zones
From top to bottom:
Throat
Stack
Belly
Bosh
Tuyere zone
Hearth

How a Blast Furnace Operates?



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Raw Materials

↓

Charging System

↓

Iron Ore + Coke + Flux

↓

Reduction in Furnace

↓

Hot Blast Through Tuyeres

↓

Combustion + Reduction

↓

Molten Iron + Slag

↓

Tapping

↓

Hot Metal → ~~Steelmaking~~

CRITICAL BLAST FURNACE ASSETS FOR PREDICTIVE MAINTENANCE

Maintenance Zone	Typical Problem	Key Parameters to Monitor
Refractory Lining	Wear / Erosion	Temperature, Heat Flux
Cooling Staves	Leakage / Blockage	Water Flow, Pressure, Temperature
Tuyeres	Damage / Blockage	Temperature, Cooling Water
Hot-Blast System	Valve / Blower Problems	Pressure, Vibration, Temperature
Blowers	Bearing Failure	Vibration, Temperature
Pumps	Bearing / Seal Failure	Vibration, Current
Motors	Overheating	Current, Temperature
Charging System	Mechanical Failure	Position, Vibration
Gas System	Leakage / Abnormal Pressure	Pressure, Gas Composition
Fans	Bearing / Impeller Problems	Vibration, Current
Valves	Sticking / Leakage	Position, Pressure

FROM REACTIVE MAINTENANCE TO PREDICTIVE MAINTENANCE

01 — REACTIVE MAINTENANCE

Failure → Stop → Inspect → Repair → Restart

Limitation:

Problem is detected after failure and damage.

02 — PREVENTIVE MAINTENANCE

Schedule → Inspect → Replace → Maintain

Limitation:

Healthy components may be replaced, while faults can develop between inspections.

03 — PREDICTIVE MAINTENANCE

Sensors → Condition Monitoring → AI/ML Analysis → Anomaly Detection → Failure Prediction → Maintenance

Advantage:

Detects developing faults before critical failure occurs.

KEY TRANSITION

“FIX AFTER FAILURE” → “PREDICT BEFORE FAILURE”

WHY IS BLAST FURNACE PREDICTIVE MAINTENANCE DIFFICULT?

BLAST FURNACE OPERATION

↓

COMPLEX OPERATING CONDITIONS

↓

MULTIPLE SENSOR & PROCESS DATA

↓

DATA UNCERTAINTY / NOISE

↓

DIFFICULT CONDITION ASSESSMENT

↓

UNCERTAIN FAILURE PREDICTION

↓

CHALLENGING MAINTENANCE DECISION

1. EXTREME ENVIRONMENT
High temperature and chemical attack

2. LIMITED DIRECT ACCESS
Critical components cannot always be inspected online

3. MULTI-PHYSICS DEGRADATION
Thermal + hydraulic + mechanical + chemical effects

4. PROCESS–EQUIPMENT COUPLING
Process abnormalities can look like equipment faults

5. DATA QUALITY
Noise, missing data, sensor drift and changing operating conditions

Therefore:
Single-sensor threshold monitoring is insufficient.

WHAT DATA DO WE COLLECT?

Thermal

- Furnace temperature
- Stave temperature
- Tuyere temperature
- Cooling-water temperature
- Hot-blast temperature
- Bearing temperature

Flow

- Cooling-water flow
- Hot-blast flow
- Gas flow
- Fuel flow

Gas/process

- CO
- CO_2
- H_2
- O_2
- Gas flow
- Hot-metal temperature
- Coke rate

Pressure

- Differential pressure
- Hot-blast pressure
- Gas pressure
- Cooling-water pressure
- Differential pressure

Mechanical

- Vibration
- Motor current
- Motor temperature
- Rotational speed
- Valve position

FAILURE MODE 1

REFRACTORY DEGRADATION

The refractory protects the furnace shell from extreme temperature and chemical attack.

Thermal + Chemical + Mechanical Stress
↓
Refractory Wear
↓
Reduced Wall Thickness
↓
Increased Heat Transfer
↓
Abnormal Thermal Pattern

What can we monitor?

- Stave temperature
- Cooling-water temperature
- Heat flux
- Thermal gradients
- Long-term temperature trends

Tata Steel has worked on online/automated methods for monitoring blast-furnace stave condition, including development of ultrasonic methods for measuring remaining stave thickness

FAILURE MODE 1

Predictive maintenance

The degradation state is inferred from measurable thermal and cooling-system behaviour

Temperature Trend

↓

AI / Statistical Model

↓

Degradation Indicator

↓

Risk Level

↓

Inspection / Maintenance

FAILURE MODE 2

COOLING SYSTEM FAILURE

Cooling is critical to maintaining the integrity of the furnace structure.

What can we monitor?

- Cooling Water Flow
- Cooling Water Pressure
- Inlet Temperature
- Outlet Temperature
- Temperature Difference
- Stave Temperature

Example:

Cooling Flow ↓
+
Pressure ↓
+
Stave Temperature ↑
↓
Abnormal Cooling Condition
↓
Warning
↓
Inspection

Predictive maintenance

Abnormal Cooling Trend
↓
Early Warning
↓
Inspection
↓
Planned Cooling System Maintenance

FAILURE MODE 3

TUYERE FAILURE / ABNORMALITY

Tuyeres are critical because hot blast enters through them.Tuyere failure can have severe consequences because cooling-water ingress and gas-related hazards can escalate rapidly.

The World Steel Association's 2025 safety recognition program describes POSCO's use of AI image processing with tuyere cameras to identify abnormalities such as blockage and tuyere bending and alert workers.

Monitoring architecture

TUYERE CAMERA

↓

IMAGE PRE-PROCESSING

↓

COMPUTER VISION

↓

ABNORMALITY DETECTION

↓

RISK CLASSIFICATION

↓

OPERATOR ALERT

↓

SAFE RESPONSE

Potential issues

- Blockage
- Damage
- Cooling problems
- Abnormal raceway conditions

Predictive maintenance

Visual / Thermal Abnormality

↓

AI Detection

↓

Early Warning

↓

Inspection

↓

Planned Intervention

FAILURE MODE 4

BLOWER / MOTOR / PUMP FAILURE

These are rotating assets

Typical failure progression

Normal

↓

Small vibration increase

↓

Bearing degradation

↓

Temperature increase

↓

Vibration increases further

↓

High failure probability

↓

Maintenance

What can we monitor?

- Vibration
- Temperature
- Motor Current
- Speed
- Pressure

FAILURE MODE 4

BLOWER / MOTOR / PUMP FAILURE

Predictive maintenance

Instead of :

Bearing Failure

↓

Blower Stops

↓

Process Disruption

We want :

Abnormal Trend

↓

Early Warning

↓

Inspection

↓

Planned Bearing Replacement

AI ROLE IN BLAST FURNACE MAINTENANCE

Stage 1 — Detect abnormal behaviourt

Normal sensor pattern
↓
Current sensor pattern
↓
Deviation
↓
ANOMALY

Possible techniques:

- Isolation Forest
- Autoencoder
- One-Class SVM

Stage 2 — Identify fault

Anomaly
↓
Classification
↓
Cooling / Bearing / Motor / Tuyere / Sensor

Possible techniques:

- Random Forest
- XGBoost
- SVM
- Neural Networks

Stage 3 — Predict degradation

Historical Trend
↓
Prediction Model
↓
Future Condition

Possible techniques:

- Digital Twin
- XGBoost
- Regression Analysis
- SVR (Support Vector Regression)

Technologies used in Predictive Maintenance

Technology	Main purpose	Example in a blast furnace
IoT / Industrial Sensors	Collect real-time condition data	Temperature, pressure, flow, vibration
Condition Monitoring	Detect abnormal equipment behavior	Cooling-water flow or tuyere temperature monitoring
Data Analytics & Signal Processing	Clean and identify patterns in sensor data	Trend analysis, filtering, feature extraction
Digital Twin	Create a virtual representation of the physical asset	physical assetSimulate furnace/ cooling-system behavior
Cloud/Edge Computing & IIoT	Process and transmit large amounts of industrial data	Real-time monitoring through SCADA/industrial networks
Machine Learning / AI	Predict faults and degradation	Predict cooling-stave or pump failure

IIoT Sensor Layer for Blast Furnace Predictive Maintenance

Blast Furnace IIoT Sensor Network

HYDRAULIC
- Cooling-water flow
- Cooling-water pressure
- Cooling-water temperature
- Tuyere cooling parameters

THERMAL
- Refractory / shell temperature
- Cooling-water temperature
- Hot-blast temperature
- Hearth temperature

PROCESS
- CO concentration
- CO_2 concentration
- H_2 concentration
- Furnace pressure
- Gas flow

MECHANICAL
- Vibration
- Motor current
- Bearing temperature
- Motor temperature

IIoT Sensor Layer for Blast Furnace Predictive Maintenance

BLAST FURNACE

↓

MULTI-SENSOR DATA

↓

PLC / DAQ / EDGE GATEWAY

↓

SCADA / HISTORIAN

↓

DATA PROCESSING

↓

AI / ML MODELS

↓

HEALTH & FAILURE PREDICTION

↓

MAINTENANCE DECISION

SENSORS

↓

Temperature | Pressure | Flow | Gas Composition | Vibration | Current

DATA PROCESSING

↓

Filtering | Synchronization | Feature Extraction

AI / ML

↓

Anomaly Detection | RUL Prediction | Failure Classification

PREDICTIVE OUTPUT

↓

Early Warning | Remaining Useful Life | Maintenance Priority

AI Integration for Blast Furnace Predictive Maintenance

Historical sensor data

↓

Training dataset

↓

Validation dataset

↓

Test dataset

↓

Model prediction

↓

Compare with actual failure / degradation

↓

Performance evaluation

OPTIMIZATION FEATURES

Sensor Fusion

combines temperature, flow, pressure,
vibration and gas data to reduce false alarms.

Edge Computing

detects critical abnormalities with low response time.

Health Index

converts multiple sensor parameters into
a simple equipment condition score.

RUL Prediction

estimates remaining useful life of
critical components.

Physics + AI

combines blast-furnace behaviour with
machine-learning predictions.

Digital Twin

simulates degradation and supports
maintenance planning.

Technologies used in Predictive Maintenance

Layer	Examples	Purpose
Control	ABB 800xA, Honeywell Experion	PLC/DCS automation
SCADA	Siemens WinCC, AVEVA	Monitoring & alarms
Data Historian	AVEVA PI, Honeywell PHD	Store & analyze process data
IIoT	OPC UA, MQTT, Industrial Edge	Connect OT data & edge devices
AI / ML	Python, MATLAB	Anomaly & failure prediction
Predictive Maintenance	Siemens Senseye, AVEVA Predictive Analytics	Asset health & degradation prediction
Digital Twin	MATLAB/Simulink, simulation platforms	Future-condition simulation
CMMS / EAM	SAP PM, IBM Maximo	Maintenance planning & work orders

Softwares used for predictive maintainance

CONTROL
ABB 800xA
Honeywell Experion

SCADA
Siemens WinCC
AVEVA

HISTORIAN
AVEVA PI
Honeywell PHD

IIoT / EDGE
OPC UA
MQTT
Industrial Edge

AI / ML
Python
MATLAB

PREDICTIVE MAINTENANCE
Senseye
AVEVA Predictive Analytics

EAM / CMMS
SAP PM
IBM Maximo

OT LAYER
↓
SCADA / DCS
↓
HISTORIAN
↓
IIoT / EDGE
↓
AI / ML
↓
PREDICTIVE MAINTENANCE
↓
CMMS / EAM

FUTURE RESEARCH

- Real-Time Data Integration: Integrate multi-sensor data from temperature, pressure, flow, vibration and vision systems.
- Advanced AI/ML Models: Develop and compare models for anomaly detection, fault classification and degradation prediction.
- Digital Twin Integration: Develop a digital representation of critical blast furnace assets for condition simulation and prediction.
- Model Validation: Validate the proposed framework using historical plant data and actual failure/degradation events.
- Predictive Maintenance Optimization: Develop risk-based maintenance recommendations considering equipment health, production requirements and maintenance cost.
- Industrial Deployment: Integrate the predictive framework with existing SCADA/DCS and CMMS/EAM systems for real-time decision support.

CONCLUSION

- Blast furnace critical assets operate under severe thermal, mechanical and chemical conditions, making early fault detection essential.
- Multi-sensor condition monitoring can provide a more comprehensive representation of equipment health than individual parameters.
- AI/ML techniques can support anomaly detection, fault identification and degradation prediction using historical and real-time data.
- The proposed framework connects sensing → data processing → AI analytics → asset health assessment → maintenance decision support.
- Future validation with real industrial data can improve prediction reliability and enable practical deployment.

Thank you